One Lyapunov Ratio, Three Global Phases: The May–Leonard Coexistence–Cycle Atlas

HCS-C358 source-local reconstruction

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Abstract

For the three-species cyclic May–Leonard competition flow in the strict intransitive chamber, we prove an all-interior-orbit classification. The normalized product $\mathcal{R} = xyz/(x+y+z)^3$ has a derivative whose sign is exactly that of $2-a-b$. Below the critical plane it yields global convergence to coexistence. On the plane it combines with an exact logistic time change to give a foliation by periodic rock–paper–scissors leaves and an elliptic period quadrature. Above the plane, the positive diagonal is the complete stable set of the unstable coexistence point, while every other interior orbit has the full oriented boundary heteroclinic cycle as its omega-limit set and has diverging residence times. Exact rational and symbolic receipts audit the identities and conventions without replacing the global proof. No literature-priority or arithmetic-target claim is made.

Revision certificate. Round-zero invariant and coexistence closure.

1 Model, chamber, and main result

For $a, b \geq 0$, consider

$$\begin{aligned} \dot{x} &= x(1 - x - ay - bz), \\ \dot{y} &= y(1 - bx - y - az), \\ \dot{z} &= z(1 - ax - by - z). \end{aligned} \tag{1}$$

We work in the strict cyclic chamber

$$(a-1)(b-1) < 0. \tag{2}$$

Put $S = x + y + z$, $Q = xy + yz + zx$, and, in the interior, $\mathcal{R} = xyz/S^3$. Let

$$p = \frac{1}{1+a+b}(1,1,1), \qquad E_1 = (1,0,0),\ E_2 = (0,1,0),\ E_3 = (0,0,1).$$

Theorem 1 (Complete cyclic-chamber atlas)**.** *The closed positive octant is forward invariant; its interior is preserved; and every positive semiorbit is bounded and forward complete. The point p is the unique interior equilibrium. For every interior initial state:*

(i) *If $a + b < 2$, the solution converges to p.*

(ii) *If $a + b = 2$, then S obeys a logistic equation. With $(u, v, w) = (x, y, z)/S$, $\delta = 1 - a = b - 1$, and $d\tau = S\,dt$, the normalized flow is*

$$u_\tau = \delta u(v - w), \quad v_\tau = \delta v(w - u), \quad w_\tau = \delta w(u - v). \tag{3}$$

Its centre is fixed, and each level $uvw = h$, $0 < h < 1/27$, is one periodic orbit. If $r_-(h) < 1/3 < r_+(h)$ are the roots in $(0,1)$ of $r(1-r)^2 = 4h$, its least τ-period is

$$T(h) = \frac{2}{|\delta|} \int_{r_-(h)}^{r_+(h)} \frac{dr}{\sqrt{r\{r(1-r)^2 - 4h\}}}. \tag{4}$$

Every original noncentral solution approaches a unique phase translate of its normalized periodic orbit.

(iii) *If $a + b > 2$, the complete interior stable set of p is the positive diagonal. Every other interior orbit has as its omega-limit set the full three-connection boundary heteroclinic cycle and spends unbounded successive times near its axial saddles. For $a < 1 < b$ its orientation is $E_1 \to E_3 \to E_2 \to E_1$; for $b < 1 < a$ the arrows reverse.*

At the closure point $a = b = 1$, all ratios are fixed and the simplex $S = 1$ consists entirely of equilibria. The separate walls $a = 1$ or $b = 1$ are nonhyperbolic and are not included in (2).

2 Positive flow and the source identity

Each coordinate in (1) is itself times a locally Lipschitz function. Thus a zero coordinate remains zero, whereas a positive one cannot vanish at finite time. Since $a, b \geq 0$,

$$\dot{x} \leq x(1-x), \qquad \dot{y} \leq y(1-y), \qquad \dot{z} \leq z(1-z).$$

Scalar comparison proves boundedness and continuation for all positive time. For some $C = C(a,b) > 0$, summing (1) also gives $\dot{S} \geq S - CS^2$. Consequently an interior semiorbit eventually lies in a compact annulus bounded away from the origin.

At an interior equilibrium,

$$\begin{pmatrix} 1 & a & b \\ b & 1 & a \\ a & b & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

The determinant is

$$(1+a+b)(1+a^2+b^2-a-b-ab),$$

and twice the second factor is $(a-b)^2 + (a-1)^2 + (b-1)^2$. It vanishes only at $a = b = 1$. Subtracting cyclic rows therefore gives the sole positive solution p in the strict chamber.

Lemma 1 (Normalized-product identity)**.** *Along every interior solution,*

$$\dot{S} = S - S^2 + (2-a-b)Q, \tag{5}$$

$$\frac{d}{dt}\log(xyz) = 3 - (1+a+b)S, \tag{6}$$

$$\frac{d}{dt}\log \mathcal{R} = (2-a-b)\frac{S^2-3Q}{S}, \tag{7}$$

$$S^2 - 3Q = \tfrac{1}{2}\{(x-y)^2 + (y-z)^2 + (z-x)^2\}. \tag{8}$$

Proof. Adding (1) and using $x^2+y^2+z^2 = S^2 - 2Q$ gives (5). Dividing each equation by its coordinate and adding gives (6). Equation (7) is (6) minus $3\dot{S}/S$, and direct expansion gives (8). ☐

The derivative in (7) is strict away from the positive diagonal whenever $a+b \neq 2$. In particular there can be no nonconstant interior periodic orbit off the critical plane.

3 Subcritical global coexistence

Assume $a+b<2$. By Lemma 1, $\mathcal{R}$ increases and is bounded above by $1/27$. The bound $\mathcal{R}(t) \geq \mathcal{R}(0) > 0$ keeps $(x/S, y/S, z/S)$ away from the simplex boundary, while the annulus estimate controls S. Hence the orbit is eventually contained in a compact subset of the interior. LaSalle's invariance principle applied to $\log \mathcal{R}$ places its omega-limit set on $x=y=z$. That diagonal is invariant, and $x=y=z=r$ reduces (1) to

$$\dot{r} = r\{1-(1+a+b)r\}. \tag{9}$$

The only compact complete diagonal orbit in the positive annulus is the equilibrium $r = (1+a+b)^{-1}$. Thus the omega-limit set is $\{p\}$ and the whole orbit converges to p, proving Theorem 1(i).